ADAPTIVE CLASSROOM ENERGY MANAGEMENT

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by

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CERTIFICATE



This is to certify that the report entitled ***ADAPTIVE CLASSROOM ENERGY MANAGEMENT*** *submitted by* ***ATHUL KRISHNA K H (PKD23EC022), FATHIMATHU SAJA K C (PKD23EC029), SALMAN FARIZ T M (PKD23EC058), SHIFANATH T S (PKD23EC061)*** *to the Department of Electronics and Communication Engineering, Government Engineering College Palakkad, Sreekrishnapuram-678633, in partial fulfilment of the requirement for the award of B-Tech Degree in Electronics and Communication Engineering, is a bonafide record of the project carried out by them under our guidance and supervision.*

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ABSTRACT

This project proposes an adaptive classroom energy management system employing ESP32 technology and sensors to dynamically control electrical appliances based on real-time environmental conditions. By integrating microcontroller capabilities, the system processes data from PIR, temperature, and light sensors efficiently, optimizing the operation of fans and lights to alleviate energy wastage. The use of sensors enables the system to respond dynamically to occupancy and ambient changes, improving overall power efficiency and maintaining classroom comfort. The implementation of this intelligent automation system showcases the potential for sensor-based solutions in enhancing sustainable energy management in educational institutions

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Chapter 1

INTRODUCTION

In modern classrooms, appliances like fans and lights are manually controlled, which often leads to significant energy wastage. To address this, an automated system using PIR, temperature, and light sensors can intelligently control these appliances by detecting student presence and environmental conditions. By leveraging an ESP32D microcontroller to process real-time sensor data, the system ensures efficient energy usage and maintains occupant comfort [1,2].

1.1 PROBLEM STATEMENT

Manual operation of classroom lights and fans without considering occupancy, temperature, and ambient conditions leads to unnecessary energy wastage, creating the need for an adaptive automated system for efficient appliance control.

1.2 OBJECTIVES

The main objectives of the proposed Adaptive Classroom Energy Management System are listed below:

- To automatically control classroom lights and fans based on real-time environmental conditions.
- To sense classroom parameters such as occupancy, temperature,

and light intensity using sensors and automatically switch appliances ON or OFF through the ESP32 microcontroller to reduce unnecessary energy consumption while maintaining basic comfort.

- To develop a simple and efficient automated system where a low-power LED light (9–12 W) and a fan (35–60 W) operate only when required.
- To display real-time classroom information such as temperature, motion detection status, light intensity, and the ON/OFF status of lights and fans using an OLED display.
- To provide manual override switches for controlling the fan and lights whenever necessary.

1.3 MOTIVATION

The motivation behind developing the Adaptive Classroom Energy Management System is to address the significant energy wastage caused by manually controlled classroom appliances, which often remain switched ON regardless of occupancy and environmental conditions. In many classrooms, lights and fans continue operating even when not required, leading to unnecessary power consumption and reduced energy efficiency.

By utilizing ESP32 microcontroller technology along with sensors such as PIR, LDR, and LM35, the project aims to create an intelligent automation system capable of adapting appliance operation in real time based on occupancy, temperature, and ambient light intensity. The system dynamically controls classroom appliances by automatically switch-

ing them ON or OFF according to environmental conditions, thereby reducing energy wastage while maintaining basic occupant comfort.

The project also aims to demonstrate how sensor-based automation and embedded systems can contribute to the development of smart and energy-efficient classrooms. Through efficient monitoring and intelligent control, the system provides a practical and cost-effective solution for sustainable energy management in educational institutions.

1.4 ORGANISATION OF REPORT

The main body of the report is preceded by detailed contents including abstract. This is followed by system description.

- Chapter 1 gives a brief introduction about the aim of the project.
- Chapter 2 provides the literature review.
- Chapter 3 gives a detailed discussion on the proposed system.
- Chapter 4 explains the requirements and specifications of hardware and software components.
- Chapter 5 presents analysis of the results and its review.
- Chapter 6 is the conclusion of the report along with the limitations and future work.

The main report is then followed by references.

Chapter 2

REVIEW OF RELATED LITERATURE

Several studies have been conducted on sensor-based automation systems for improving energy efficiency in classrooms and indoor environments. These studies mainly focus on reducing unnecessary power consumption through automatic control of electrical appliances using sensors and microcontrollers. Some of the important studies related to the proposed project are discussed below.

The research paper titled "Implementation of Automated Smart Classroom Using Sensors" (IJIAREC, 2018) proposed a system for automating classroom appliances based on real-time environmental conditions. The system utilized sensors to detect human presence and automatically control appliances such as lights and fans. Temperature sensors were also used for fan control to regulate fan operation for maintaining user comfort [1]. for maintaining user comfort. The study demonstrated that automation can effectively reduce manual operation and improve energy efficiency in classroom environments. However, the paper identified certain limitations such as the lack of detailed energy-saving analysis and limited coverage area, which makes the system less suitable for large classrooms or halls. These limitations indicate the need for a more scalable and efficient automation system .

Another important study titled “Energy Saving System Using a PIR Sensor” (IJRASET, 2023) focused on reducing energy consumption using PIR sensor-based occupancy detection [1]. The system automatically switched appliances ON and OFF depending on human presence, thereby minimizing unnecessary power usage. Relay modules controlled through a microcontroller were used for appliance operation. Although the system improved convenience and energy efficiency, it also had several limitations. The system could not determine the number of occupants in a room and operated with a fixed delay response, which reduced efficiency in certain situations. Moreover, the absence of ambient light sensing caused lights to remain ON even when sufficient natural light was available. These drawbacks highlighted the importance of integrating additional environmental sensing techniques for better intelligent control .

The reviewed literature clearly shows that sensor-based automation systems play an important role in reducing energy wastage and improving appliance control efficiency [1;2;3].

Most of the existing systems mainly depend on occupancy detection alone, which limits their ability to make intelligent decisions under varying environmental conditions. In addition, many existing systems use simpler microcontrollers with limited processing capability and scalability.

To overcome these limitations, the proposed Adaptive Classroom Energy Management System integrates multiple sensors such as PIR, LDR, and LM35 along with the Espressif Systems ESP32D microcontroller. The inclusion of the LDR sensor enables intelligent light control based on ambient light intensity, while the temperature sensor supports

efficient fan control according to room temperature. The ESP32D microcontroller provides improved processing capability, better scalability, and built-in Wi-Fi support for future IoT-based extensions.

Another significant improvement in the proposed system is the inclusion of an OLED display for real-time monitoring of classroom conditions such as temperature, light intensity, motion detection status, and appliance ON/OFF condition. This enhances user interaction and system transparency. Furthermore, the multi-sensor approach adopted in the proposed project ensures that appliances operate only when required, thereby improving energy conservation while maintaining basic user comfort.

Overall, the literature survey indicates that existing classroom automation systems have made considerable progress in energy management and intelligent appliance control. However, limitations related to scalability, environmental adaptability, and monitoring capabilities still exist. The proposed Adaptive Classroom Energy Management System addresses these issues through multi-sensor integration, real-time monitoring, and the use of a more advanced microcontroller platform, resulting in a more efficient, flexible, and practical solution for smart classroom applications.

Chapter 3

PROPOSED SYSTEM

This includes the block diagram, its detailed explanation, flow chart and its explanation.

3.1 BLOCK DIAGRAM

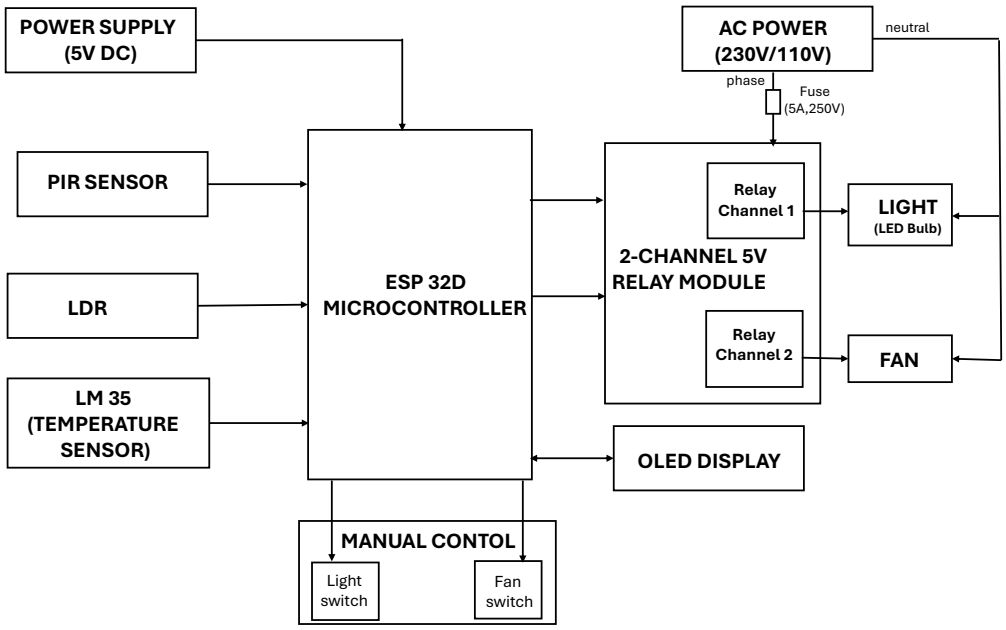


Figure 3.1: Block Diagram

The system architecture for the Adaptive Classroom Energy Management system is designed around a centralized microcontroller that interfaces with various input sensors, control mechanisms, and output peripherals. The primary objective of this architecture is to automate the operation of classroom appliances (lights and fans) based on real-time environmental factors and occupancy status, thereby optimizing

energy efficiency.

System Components and Interconnections

The block diagram illustrated in **Figure 3.1** can be broadly classified into four major functional modules: the Power Supply, Input Module (Sensors and Manual Overrides), Core Processing Unit, and Output Module (Actuators and Display).

- **Power Supply:** A regulated **5V DC Power Supply** provides the stable operating voltage required by the ESP32D microcontroller and the associated sensor/relay modules.
- **Core Processing Unit:** The **ESP32D Microcontroller** acts as the central brain of the system. It is responsible for continuously polling data from the input sensors, executing the control logic, handling manual override interrupts, and generating corresponding actuation signals for the outputs.
- **Input Module (Sensors & Manual Control):**
  - **PIR Sensor:** Detects human motion within the classroom to determine real-time occupancy.
  - **LDR (Light Dependent Resistor):** Measures the ambient light intensity to evaluate if artificial illumination is required.
  - **LM35 Temperature Sensor:** monitors the ambient room temperature to determine the necessity of fan operation.
  - **Manual Control Switches:** Provides user-accessible physical switches for lights and fans, allowing occupants to manually

override the automated system logic when necessary, ensuring system flexibility.

- **Output Module (Actuators & Displays):**
  - **2-Channel 5V Relay Module:** Acts as an electrically operated switch that provides galvanic isolation between the low-voltage DC microcontroller circuit and the high-voltage AC mains. **Relay Channel 1** controls the **Light (LED Bulb)**, while **Relay Channel 2** controls the **Fan**.
  - **OLED Display:** Provides a localized, real-time visual interface displaying the current environmental parameters (temperature, light levels) and the operational status of the appliances.
  - **AC Power Architecture:** A **230V/110V AC Power** line delivers energy to the appliances. For safety and overcurrent protection, a **Fuse (5A, 250V)** is integrated inline with the phase wire before routing through the relay channels to the loads.

Functional Workflow

The operational flow of the system begins with the sensors capturing ambient data from the classroom environment. The ESP32D processes these inputs using predefined threshold algorithms. For instance, if the PIR sensor detects occupancy and the LDR senses low ambient light, the microcontroller sends a trigger signal to Relay Channel 1 to turn on the LED light. Similarly, if the temperature recorded by the LM35 exceeds a set threshold during occupied hours, Relay Channel 2 is energized to turn on the fan.

Conversely, if no motion is detected over a designated timeout period, the microcontroller automatically de-energizes the relays, shutting down all appliances to prevent idle power wastage. Throughout this process, system statuses are dynamically updated on the OLED display, while the manual control unit remains available to handle immediate user preferences.

3.2 FLOWCHART

The operational logic and decision-making process of the Adaptive Classroom Energy Management system are structurally mapped out in the flowchart shown in **Figure 3.2**. The control algorithm is designed as a continuous, closed-loop firmware sequence executed by the ESP32D microcontroller to balance occupant comfort with optimized power consumption.

Initialization and Data Acquisition

The sequence initiates at the **Start** block upon powering up the system.

- **System Initialization:** The microcontroller first configures its General Purpose Input/Output (GPIO) pins, establishes communication protocols with the OLED display, and calibrates the connected sensors (PIR, LM35, and LDR).
- **Read Sensors Data:** Once initialization is complete, the system enters its primary operational loop, continuously polling real-time environmental data. This includes human motion detection from the PIR sensor, ambient light intensity from the LDR, and ambient room temperature from the LM35 thermal sensor.

Decision Logic and Control Sequences

The core firmware processes the acquired data through a multi-tier conditional check to execute the appropriate load management commands:

1. **Manual Override Evaluation:** The algorithm immediately verifies the status of the physical override switches (**Is Manual Control Overridden?**).
    - **YES Branch:** If user intervention is detected, the automated sensor logic is bypassed (**Bypass Automation Logic**). The system directly executes the physical user commands (**Execute Manual Switching Commands**) to toggle the lights or fans, ensuring that human preference retains ultimate priority.
    - **NO Branch:** If no manual override is active, the system transitions to its fully automated energy-saving algorithms.
2. **Occupancy Verification:** The system evaluates the PIR sensor state (**Is Room Occupied?**).
    - **NO Branch:** If no motion is detected within a designated timeframe, the classroom is deemed empty. The system triggers the relay module to cut off mains power to both loads (**Turn off all AC Appliances**), mitigating passive energy wastage.
    - **YES Branch:** If the room is occupied, the algorithm splits into parallel evaluation tracks to independently manage the lighting and climate sub-systems.
3. **Lighting Automation Sub-loop:** The microcontroller checks

the LDR values against a preset lux threshold (**Is Light Intensity Below Threshold?**).

- If **YES** (insufficient natural light), a command is issued to **Turn ON the Light** via Relay Channel 1.
- If **NO** (adequate natural light available), it triggers the system to **Turn OFF the Light**.

4. **Climate Automation Sub-loop:** Simultaneously, the system compares the LM35 thermal data against the comfort threshold (**Is Temperature Above Threshold?**).

   - If **YES** (high ambient temperature), the microcontroller acts to **Turn ON the Fan** via Relay Channel 2.
   - If **NO** (cool ambient temperature), it defaults to **Turn OFF the fan**.

Status Update and Loop Continuity

Regardless of whether the system operated via manual override, automated sensor tracking, or an idle state shutdown, all diagnostic pathways converge at the final processing stage:

- **Update OLED Display Status:** The microcontroller refreshes the localized visual display, outputting current temperature readings, sensor values, and active load parameters for user awareness.
- **Loop Feedback:** Following the display refresh, the algorithm executes a feedback loop, routing back to the **Read Sensors Data** block. This closed-loop design ensures that the entire system re-

mains dynamic, continuously adapting to real-time institutional environment modifications.

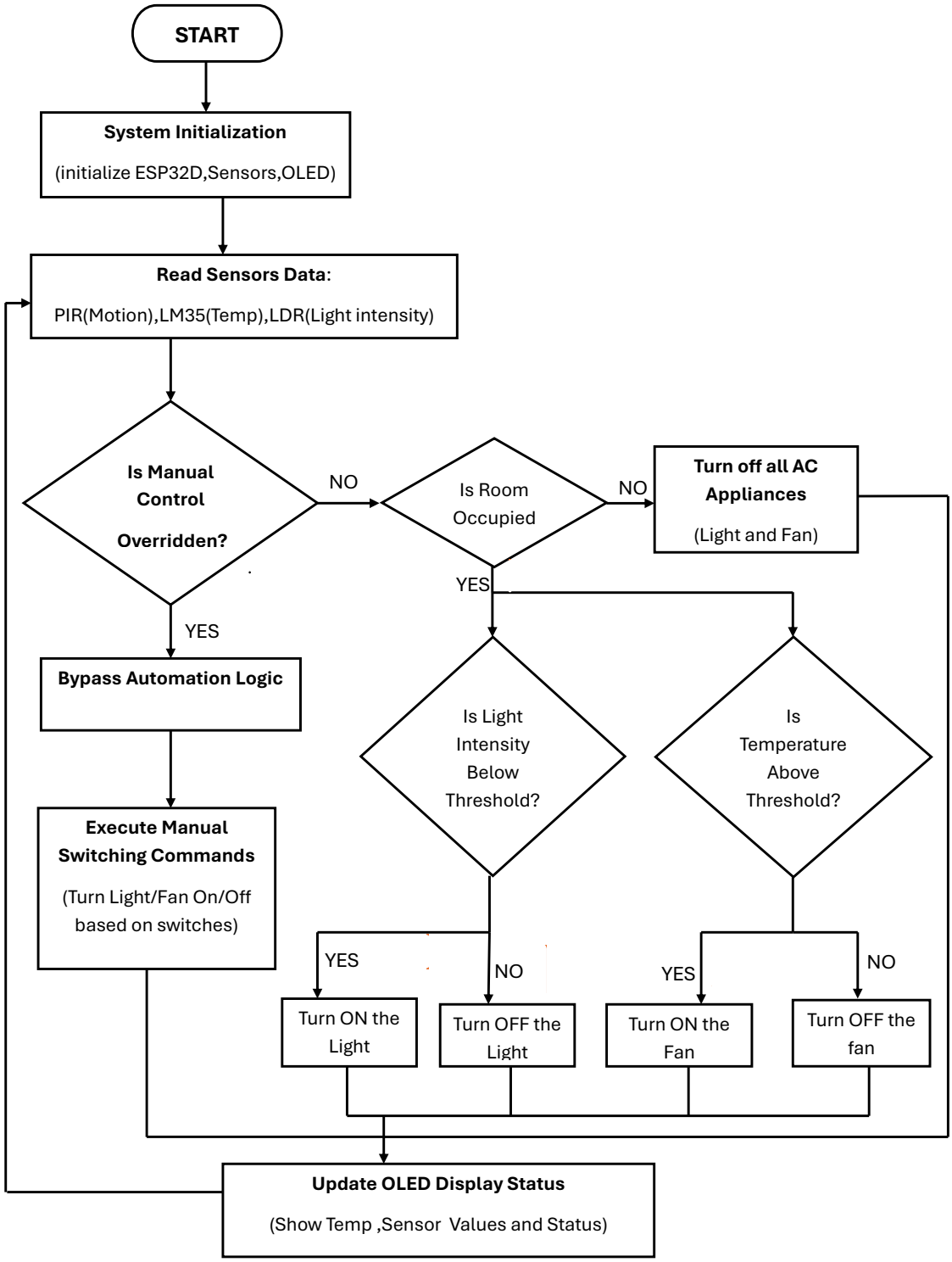


Figure 3.2: Flowchart of the Proposed System

Chapter 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE

4.1.1 ESP32D MICROCONTROLLER



Figure 4.1: ESP32D

The ESP32D is a high-performance, low-power System-on-Chip (SoC) that offers significant advantages in the development of smart environments and IoT applications. It features a dual-core 32-bit Xtensa LX6 microprocessor, capable of operating at speeds up to 240 MHz. Developed by Espressif Systems, the ESP32D provides a flexible embedded processing platform that allows users to implement custom control logic tailored to specific automation requirements. It is widely used in diverse

fields, including home automation, wireless monitoring, and industrial control.

The ESP32D consists of a rich set of peripheral interfaces, including ADC, DAC, SPI, and I2C, interconnected via programmable internal buses. These devices offer high flexibility and parallel processing capabilities, enabling the real-time management of multiple sensors and actuators. With integrated Wi-Fi and Dual-mode Bluetooth, the ESP32D facilitates seamless integration with external networks and communication interfaces, allowing for remote monitoring of classroom data.

Key features of the ESP32D include on-chip memory resources and dedicated hardware for low-power operation, ensuring that the controller itself remains energy-efficient. In this project, the ESP32D serves as the central processing unit, receiving input from occupancy and environmental sensors. By executing intelligent algorithms, it ensures the precise synchronization of fan and light operations, providing a robust technical foundation for reducing power consumption in educational buildings [4].

4.1.2 PIR SENSOR



Figure 4.2: HC-SR501 PIR SENSOR

Passive Infrared (PIR) sensors are widely used in automation systems for detecting human presence and motion without physical contact. These sensors operate based on the detection of infrared radiation emitted by warm objects such as the human body. Unlike active sensors, PIR sensors do not emit radiation; instead, they sense changes in infrared energy in their surroundings.

The PIR sensor consists of a pyroelectric sensor that detects infrared radiation and converts it into an electrical signal. It also includes a Fresnel lens that helps in focusing infrared signals from a wide detection area onto the sensor. When a person moves within the detection range, the infrared radiation pattern changes, and this variation is detected by the sensor, producing a digital output signal.

The HC-SR501 PIR sensor provides a HIGH output when motion is detected and a LOW output when no motion is present. It typically operates at 5V and has an adjustable detection range of approximately 3 to 7 meters with a detection angle of about 110° to 140°. The sensor also allows adjustment of sensitivity and delay time, making it suitable for different environments and applications.

PIR sensors offer several advantages, including low power consumption, high reliability, and ease of use. They are commonly used in security systems, automatic lighting systems, and energy management applications. In the Adaptive Classroom Energy Management System, the PIR sensor plays a crucial role in detecting classroom occupancy. Based on the presence or absence of occupants, the system automatically controls appliances such as lights and fans, thereby reducing unnecessary energy consumption [7].

Furthermore, PIR sensors are relatively unaffected by visible light

conditions, making them suitable for indoor environments like classrooms. They can be easily interfaced with microcontrollers such as ESP32, enabling efficient real-time processing and automation. This makes the HC-SR501 an essential component in implementing intelligent and energy-efficient systems.

4.1.3 TEMPERATURE SENSOR(LM35)

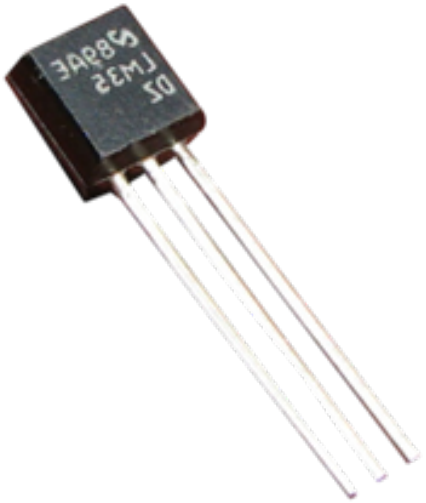


Figure 4.3: LM 35 TEMPERATURE SENSOR

The LM35 temperature sensor is widely used in electronic systems for measuring temperature accurately and efficiently. It is a precision integrated-circuit sensor whose output voltage is directly proportional to the temperature in degree Celsius. Unlike many other temperature sensors, the LM35 does not require any external calibration and provides a linear output, making it simple and reliable for practical applications. The sensor operates by converting temperature variations into corresponding analog voltage signals, which can be easily read by microcontrollers.

The LM35 consists of a sensing element and signal conditioning cir-

cuitry that produces an output voltage of 10mV per°C. For example, at 25°C, the output voltage will be 250 mV. This linear relationship allows easy conversion of voltage into temperature values. The sensor operates over a wide temperature range, typically from 55°C to +150°C, and requires a supply voltage between 4V and 30V, making it suitable for various embedded applications [6].

The LM35 offers several advantages, including high accuracy, low self-heating, and low cost. It provides reliable temperature measurements without the need for complex signal processing circuits. Due to its analog output, it can be easily interfaced with microcontrollers such as ESP32 using ADC (Analog-to-Digital Converter) pins. This makes it highly suitable for real-time monitoring systems.

In the adaptive classroom energy management system, the LM35 sensor plays a crucial role in controlling the operation of fans. The system continuously monitors the classroom temperature and compares it with a predefined threshold value. When the temperature exceeds the set limit, the fan is automatically turned ON, ensuring thermal comfort while avoiding unnecessary energy consumption. When the temperature is below the threshold, the fan remains OFF, contributing to efficient energy usage.

Furthermore, the LM35 sensor is stable under varying environmental conditions and provides consistent performance. Its simple interfacing, accurate output, and low power consumption make it an ideal choice for automation systems such as the proposed energy management system

4.1.4 LDR SENSOR (GL5528)

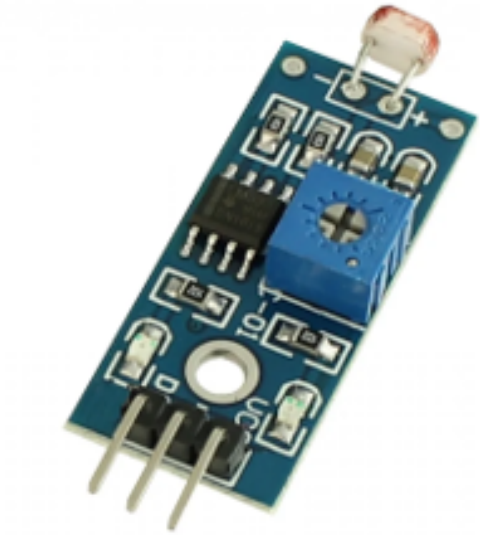


Figure 4.4: GL5528 LDR MODULE

A Light Dependent Resistor (LDR), specifically the GL5528 model, is a passive electronic component whose resistance varies depending on the intensity of light falling upon its surface. It is categorized as a photo-conductive cell because its operation is based on the principle of photoconductivity. The sensor is constructed from a high-resistance semiconductor material, typically Cadmium Sulfide (CdS), which is deposited in a zig-zag pattern on a ceramic substrate. This material contains very few free electrons when in total darkness, resulting in a very high "dark resistance" that can reach several Mega-ohms ($M\Omega$). When light photons strike the semiconductor track, they provide enough energy to release electrons from their chemical bonds, creating a surge of charge carriers that causes the electrical resistance to drop dramatically, often down to a few hundred ohms (Ω) in bright light.

The 3-pin module version of the GL5528 enhances the basic two-terminal component by integrating it into a pre-configured circuit board

designed for precision and ease of use. This module typically features a built-in voltage divider circuit and often includes an onboard potentiometer and a comparator IC such as the LM393. The three pins consist of VCC for power input (usually 3.3V to 5V), GND for common ground, and a Signal/Output pin. This configuration allows the sensor to provide a variable analog voltage output that is directly proportional to the light intensity.

Because of its high sensitivity to the visible light spectrum, the GL5528 is exceptionally effective at mimicking the human eye's perception of brightness, making it a standard choice for applications requiring automated light detection and measurement. In the Adaptive Classroom Energy Management System, the LDR sensor is used to monitor ambient light intensity and automatically control classroom lighting, thereby reducing unnecessary power consumption and improving energy efficiency [8,9].

4.1.5 OLED DISPLAY



Figure 4.5: OLED DISPLAY

OLED (Organic Light Emitting Diode) displays are widely used in electronic systems for displaying information due to their clarity, low

power consumption, and compact design. A 128×64 pixel OLED display can show text, numbers, and simple graphics using a grid of 128 horizontal and 64 vertical pixels. Unlike LCDs, OLED displays do not need a backlight because each pixel produces its own light, resulting in high contrast and better visibility even in dark environments. These displays typically use communication methods like I2C, which reduces wiring and makes it easy to connect with microcontrollers such as ESP32.

OLED displays offer advantages like fast response time, wide viewing angles, and energy efficiency since only active pixels consume power. They are lightweight and suitable for embedded and automation systems. In the Adaptive Classroom Energy Management System, the OLED display is used to show real-time data such as temperature, light intensity, motion detection, and the ON/OFF status of appliances. This helps users easily monitor system operation and improves overall usability. Additionally, OLED displays are simple to interface and program, making them a reliable choice for smart electronic projects [5].

4.1.6 2-Channel 5V RELAY MODULE

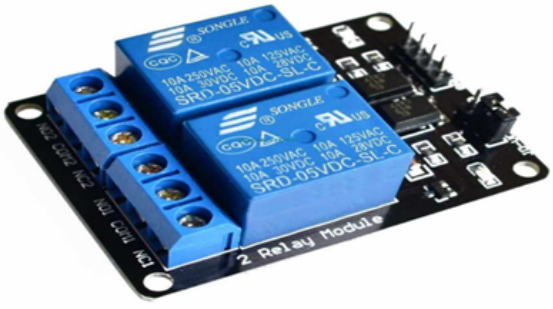


Figure 4.6: RELAY MODULE

A 2-Channel 5V Relay Module is an electromechanical switching

device designed to allow low-power electronic signals to control two independent high-voltage or high-current circuits. The module acts as an electrically operated switch, providing a safe interface between sensitive control logic and heavy electrical loads. Each of the two relays on the board is typically rated to handle up to 10A at 250V AC or 30V DC, making the module capable of managing a wide variety of standard electrical appliances.

The internal mechanism of the relay operates on the principle of electromagnetism. Each channel contains an induction coil wrapped around an iron core. When a 5V signal is applied to the input, current flows through the coil, creating a magnetic field that moves a physical metallic contact, known as the armature. This movement mechanically toggles the connection between the output terminals. A key feature of this module is galvanic isolation, which ensures that the high-voltage side is physically and electrically separated from the low-voltage control side. This protection is often reinforced with optocouplers, which use light to transmit the trigger signal, preventing electrical noise or surges from feeding back into the control electronics.

For connectivity, each relay channel is equipped with a set of three high-voltage terminals: Common (COM), Normally Open (NO), and Normally Closed (NC). In the Normally Open state, the circuit is broken until the relay is energized; in the Normally Closed state, the circuit is complete by default and breaks only when the relay is triggered. The module also includes auxiliary components such as flyback diodes to protect against voltage spikes generated by the coil's magnetic field and status LEDs that provide visual confirmation when a specific channel is active. This combination of mechanical switching and electronic

protection makes the relay module a fundamental component for power management and industrial automation [7].

4.1.7 5A FUSE



Figure 4.7: 5A Fuse

A 5A fuse is a protective electrical component used to safeguard circuits and connected devices from overcurrent conditions. It operates on a simple principle: when the current flowing through the circuit exceeds the rated value (in this case, 5 amperes), the fuse element melts and breaks the circuit, thereby stopping the flow of electricity. This prevents damage to electrical components such as microcontrollers, sensors, relay modules, and appliances. Fuses are typically made of a thin metal wire enclosed in a protective casing, designed to melt quickly when excessive current flows, ensuring fast response and reliable protection.

Fuses offer several advantages, including simplicity, low cost, and high reliability in protecting electrical systems. In the adaptive classroom energy management system, the 5A fuse plays a crucial role in ensuring safety by protecting the circuit from short circuits and overload conditions. Since the system controls AC appliances like fans and

lights through relays, there is a risk of overcurrent due to faults or improper connections. The inclusion of a fuse ensures that any abnormal current is immediately interrupted, preventing damage to components and reducing the risk of electrical hazards. Additionally, fuses are easy to install and require minimal maintenance, making them an essential safety component in such automation systems .

4.1.8 SWITCH



Figure 4.8: SWITCH

Manual switches are simple yet essential components used in electrical and electronic systems to provide direct user control over devices. A manual switch operates by physically opening or closing an electrical circuit, thereby controlling the flow of current. When the switch is in the ON position, the circuit is closed and current flows, allowing the connected appliance to operate. Conversely, when the switch is in the OFF position, the circuit is open, stopping the flow of current and turning the device off. Manual switches are widely used due to their simplicity, reliability, and ease of operation, making them a fundamental part of both domestic and industrial electrical systems.

In the adaptive classroom energy management system, manual switches

are incorporated to provide flexibility and user override capability. While the system primarily operates automatically based on sensor inputs such as occupancy, temperature, and light intensity, the manual switch allows users to control the fan and light whenever required. This ensures that the system remains practical in real-world situations where automatic operation may not always meet user preferences. Additionally, manual switches enhance safety and reliability by allowing immediate control during system faults or maintenance, ensuring uninterrupted and secure operation of classroom appliances.

4.2 SOFTWARE

4.2.1 ARDUINO IDE



Figure 4.9: Arduino IDE logo

The Arduino Integrated Development Environment (IDE) is a cross-platform, open-source software application used for writing, compiling, and uploading code to various microcontroller boards. It serves as a comprehensive suite for embedded systems development, supporting a programming language based on C and C++. The IDE is designed to simplify the process of hardware programming by providing a user-friendly interface that handles the complexities of machine-level code translation and hardware communication.

The software features a powerful Compiler that transforms human-readable source code into binary machine code that microcontrollers can execute. It includes a built-in Library Manager, which allows users to easily integrate pre-written code modules for a wide variety of sensors, displays, and communication protocols. Additionally, the IDE provides a Serial Monitor and Serial Plotter, which act as essential diagnostic tools. These features allow for real-time data exchange between the

computer and the microcontroller, enabling developers to monitor variables and visualize data streams for debugging purposes.

Modern versions of the Arduino IDE include advanced development features such as Auto-completion, syntax highlighting, and integrated debugging support. Through the use of a Board Manager, the environment can be expanded to support a diverse range of hardware architectures beyond standard Arduino boards, including specialized 32-bit processors and IoT-enabled chips. Compatible with Windows, macOS, and Linux, the Arduino IDE remains an industry-standard tool for rapid prototyping and educational engineering due to its balance of simplicity and professional-grade functionality [11].

4.2.2 WOKWI SIMULATOR



Figure 4.10: Wokwi logo

Wokwi is a modern, browser-based hardware simulator designed for the real-time design, testing, and debugging of embedded systems and IoT applications. It provides a comprehensive virtual environment that allows developers to simulate advanced microcontrollers, such as the ESP32 and Arduino, along with a wide array of digital and analog components. By executing actual firmware code in a high-fidelity simulation engine, Wokwi accurately mimics the behavior of physical hardware, including complex communication protocols like I2C, SPI, and

UART. This enables designers to verify their logic and hardware configurations without the need for physical components or breadboarding [13].

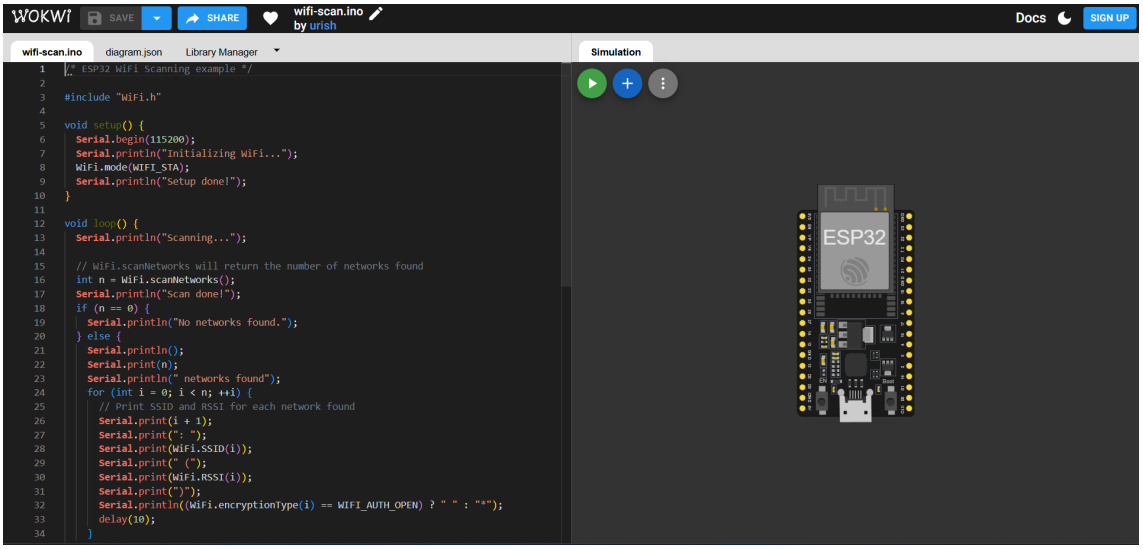


Figure 4.11: Wokwi window

Chapter 5

RESULTS AND DISCUSSION

5.1 SENSOR DETECTION AND LOGIC VERIFICATION

The simulation was conducted using the Wokwi Simulator to verify the automation logic of the ESP32. In the simulation environment, a DS18B20 temperature sensor was used along with the PIR sensor and LDR to monitor environmental conditions and ensure accurate appliance switching and OLED status reporting [13]. In the hardware prototype, the LM35 temperature sensor was used for temperature sensing.

5.2 SIMULATION OBSERVATIONS

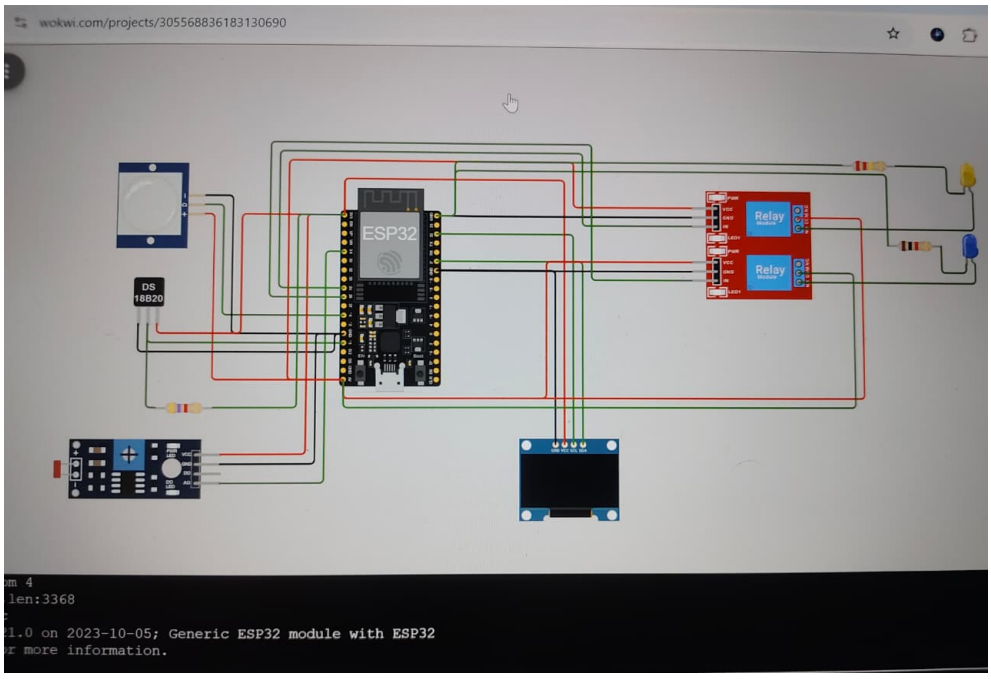


Figure 5.1: Simulation using Wokwi

Observation 1: Room Unoccupied When the PIR sensor state is LOW (No motion), the system triggers energy-saving mode. Regardless of temperature or light levels, the ESP32 turns OFF both the Yellow LED (Light) and the Blue LED (Fan). The OLED display "Occupancy: NO","LIGHT: OFF" and "Fan:OFF"

Observation 2: Room Occupied with Low Light and Low Temperature When the PIR sensor detects motion and the LDR detects low light intensity (below threshold), but the DS18B20 detects a low temperature (below threshold). The ESP32 turns ON the Yellow LED (Light) while keeping the Blue LED (Fan) OFF. The OLED display "Occupancy: YES","Light:ON" and "FAN: OFF"

Observation 3: Room Occupied with Low Light and High Temperature When the PIR sensor detects motion and the LDR detects low light intensity (below threshold) while the DS18B20 detects a high temperature (above threshold). The ESP32 turns ON both the Yellow LED (Light) and the Blue LED (Fan) . The OLED display "Occupancy: YES","Light:ON" and "Fan:ON"

Observation 4: Room Occupied with Sufficient Daylight and Low Temperature When the PIR sensor detects motion but the LDR indicates sufficient daylight (above threshold) and the DS18B20 detects a low temperature (below threshold), the ESP32 keeps both the Yellow LED (Light) and the Blue LED (Fan) OFF. The OLED displays "Occupancy: YES", "Light: OFF", and "Fan: OFF"

Observation 5: Room Occupied with Sufficient Daylight and High Temperature When the PIR sensor detects motion and the LDR indicates sufficient daylight (above threshold), but the DS18B20 detects a high temperature (above threshold), the ESP32 keeps the Yellow LED (Light) OFF while turning ON the Blue LED (Fan). The OLED displays "Occupancy: YES", "Light: OFF", and "Fan: ON"

5.3 OBTAINED SCHEMATIC

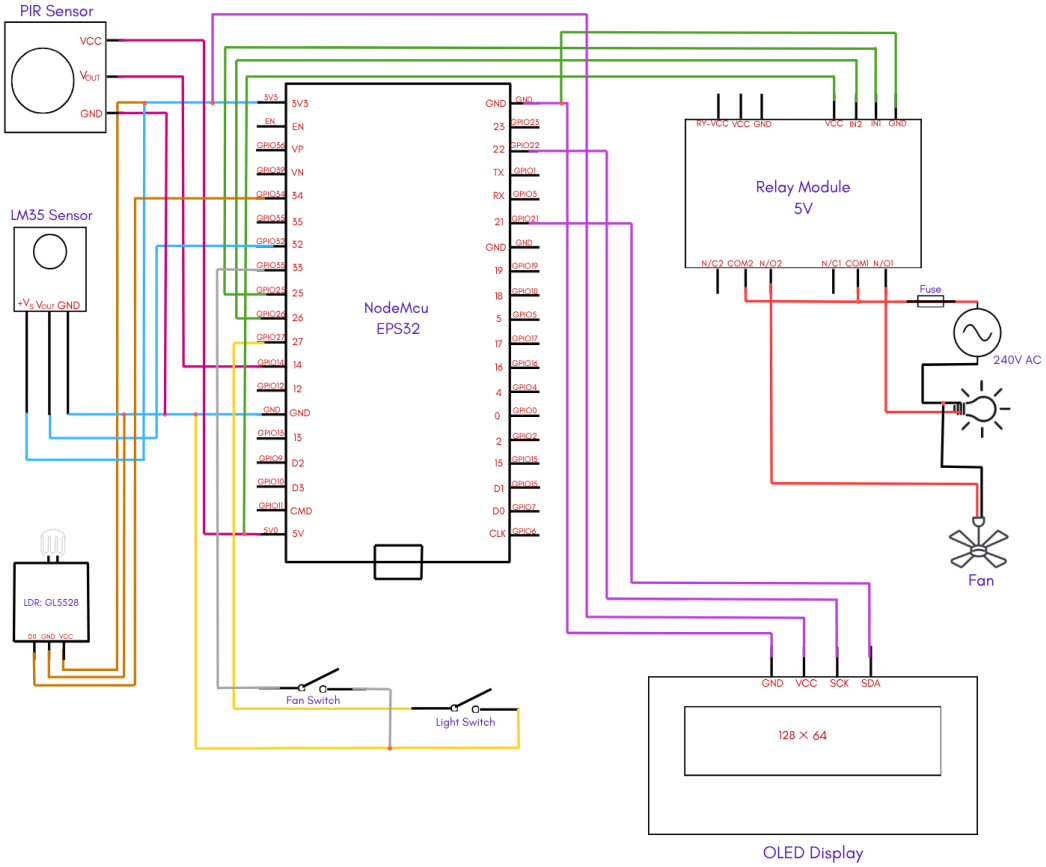


Figure 5.2: Schematic and Wiring Diagram

Figure 5.2 shows the schematic obtained after successful simulation of our design. As per the schematic, the NodeMCU ESP32 serves as the central controller where the PIR sensor, LM35 temperature sensor, and LDR are connected as inputs to monitor occupancy and environmental

conditions. Two manual override switches are also interfaced to the GPIO pins to allow for user control. The controller processes these inputs and produces the corresponding output through a 2-channel relay module to toggle the high-voltage AC light and fan circuits. Additionally, an OLED display is connected via I2C pins to provide real-time status updates, while a fuse is integrated into the power line for circuit protection.

5.4 PROTOTYPE

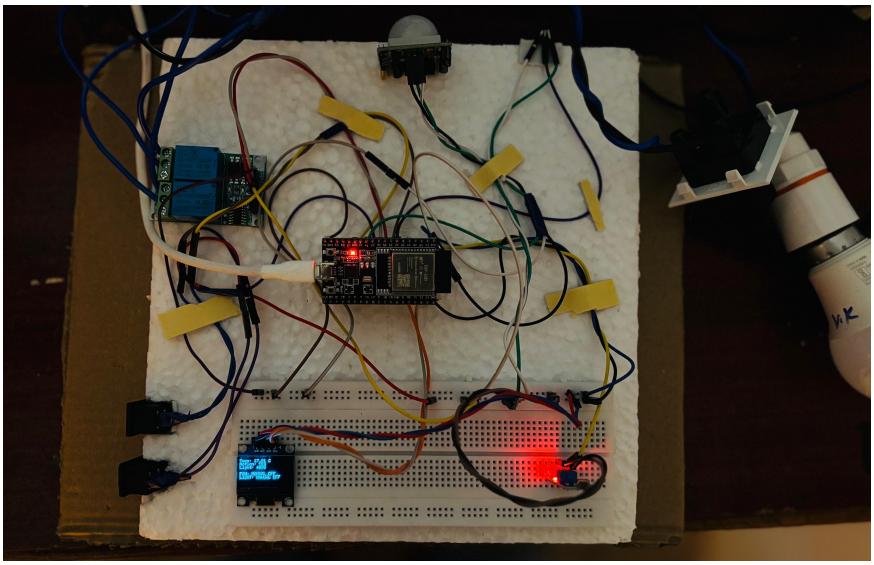


Figure 5.3: Prototype1

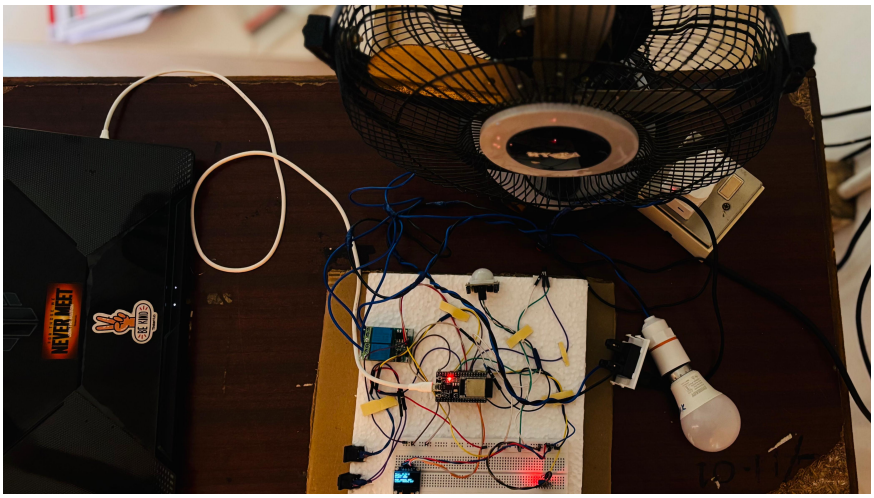


Figure 5.4: Prototype2

We developed a functional prototype of the proposed system, as shown in Figure 5.4. **ESP32-based control unit** integrated with PIR, LM35, and LDR sensors, along with a relay module to control a light and fan. An OLED display is used to monitor real-time parameters such as temperature, light intensity, occupancy status, and appliance conditions. The components are neatly assembled on a breadboard with proper wiring, along with manual switches for user control. The prototype demonstrates an efficient and professional implementation of

adaptive energy management in a classroom environment.

5.5 RESULT

The Adaptive Classroom Energy Management System was successfully designed and implemented using the ESP32 microcontroller along with PIR, LM35, and LDR sensors. The system automatically controlled classroom appliances such as lights and fans based on occupancy, temperature, and ambient light conditions. Both simulation and hardware testing confirmed the proper functioning of the automation logic.

The PIR sensor effectively detected human presence inside the classroom and enabled automatic switching of appliances only when occupancy was detected. The LDR sensor monitored ambient light intensity and controlled the lighting system efficiently by turning the lights ON only when sufficient natural light was unavailable. Similarly, the LM35 temperature sensor continuously monitored room temperature and activated the fan whenever the temperature exceeded the predefined threshold value.

The OLED display successfully provided real-time monitoring information, including temperature readings, motion detection status, light intensity, and the ON/OFF status of the fan and light. Manual override switches also functioned correctly, allowing users to manually control the appliances whenever required.

The system was first verified using the Wokwi simulation platform, where the automation logic and sensor responses were tested under different environmental conditions. After successful simulation, the hardware prototype was implemented and tested practically. The results

demonstrated that the system effectively reduces unnecessary energy consumption while maintaining user comfort inside the classroom.

Overall, the proposed system achieved reliable automated control, improved energy efficiency, reduced power wastage, and demonstrated the practical application of sensor-based smart classroom automation.

5.6 LIMITATIONS

Although the proposed Adaptive Classroom Energy Management System performs effectively, certain limitations exist in the current implementation.

The system depends heavily on the accuracy and reliability of sensors. The PIR sensor may fail to detect occupants who remain stationary for a long period, which can cause appliances to switch OFF even when the room is occupied. Similarly, LDR sensor readings can be affected by external light sources such as direct sunlight, resulting in improper light control. The LM35 temperature sensor measures temperature only at a specific point and may not accurately represent the temperature of the entire classroom.

Another limitation is that the system operates using predefined threshold values for temperature and light intensity. These fixed values may not be suitable for all classroom environments and varying weather conditions. The system also lacks adaptive learning capability, meaning it cannot automatically optimize its operation based on usage patterns.

The present system is designed mainly for small-scale classroom applications and may require additional modifications for large classrooms or multi-room installations. Furthermore, the relay module used for

switching appliances contains mechanical contacts, which may wear out over time due to repeated operation.

Although manual switches provide flexibility for users, excessive manual overriding can reduce the overall energy-saving efficiency of the system. In addition, the current implementation does not include IoT-based remote monitoring or cloud data storage features, limiting its scalability and advanced smart management capabilities.

Despite these limitations, the system provides a simple, cost-effective, and efficient solution for classroom energy management and serves as a strong foundation for future improvements in smart automation systems.

5.7 OVERCOME LIMITATIONS

The limitations of the proposed Adaptive Classroom Energy Management System can be reduced by improving both the hardware and software design of the system. More advanced and accurate sensors can be used to improve reliability and overall system performance. For example, replacing the PIR sensor with a microwave or radar-based presence sensor can help detect stationary occupants more effectively, preventing unnecessary switching OFF of appliances. Similarly, using multiple temperature sensors at different locations inside the classroom can provide more accurate environmental monitoring.

The effect of external light variations on the LDR sensor can be minimized through proper sensor placement and calibration. Advanced digital light sensors with higher accuracy can also be used instead of conventional LDR modules. In addition, adaptive threshold algorithms

can be implemented so that the system automatically adjusts light and temperature limits according to environmental conditions rather than depending on fixed values.

The scalability of the system can be improved by integrating IoT technology and wireless communication features available in the ESP32 microcontroller. This would enable remote monitoring and control through mobile applications or web-based dashboards, making the system suitable for large classrooms and smart campus environments. Cloud-based data storage and analytics can also be incorporated to monitor energy usage patterns and improve system efficiency over time.

Mechanical relay modules can be replaced with solid-state relays to improve durability, reduce switching noise, and increase system reliability. Overall energy efficiency can be further enhanced by implementing machine learning or artificial intelligence techniques that learn classroom usage patterns and optimize appliance control automatically.

Additionally, the system can be expanded by integrating extra environmental sensors such as humidity, air quality, or CO sensors to create a more intelligent and comfortable classroom environment. These improvements can make the proposed system more reliable, scalable, energy-efficient, and suitable for practical real-world applications in smart buildings and educational institutions.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The Adaptive Classroom Energy Management System developed using the ESP32 microcontroller successfully demonstrates an intelligent and energy-efficient approach for controlling classroom appliances. By integrating PIR, LM35, and LDR sensors, the system automatically manages lights and fans based on occupancy, temperature, and ambient light conditions. This automation helps reduce unnecessary power consumption while maintaining a comfortable classroom environment.

The system was tested through both simulation and hardware implementation, and the obtained results confirmed the proper functioning of the automation logic. The PIR sensor effectively detected occupancy, the LDR sensor controlled lighting according to surrounding brightness, and the LM35 monitored room temperature for automatic fan operation. The OLED display provided real-time monitoring information, while manual switches allowed user override whenever required.

The proposed system offers a simple, low-cost, and practical solution for improving energy efficiency in educational institutions. It reduces human effort, minimizes energy wastage, and demonstrates the effec-

tive application of sensor-based automation using embedded systems technology. The project also highlights the potential of smart energy management systems in supporting sustainable and intelligent infrastructure development.

6.2 FUTURE SCOPE

The Adaptive Classroom Energy Management System can be further enhanced with several advanced features to improve its performance, scalability, and intelligence. One important future improvement is the integration of IoT technology, which would allow remote monitoring and control of classroom appliances through mobile applications or web platforms. This would make the system more suitable for smart classrooms and smart campus environments.

Machine learning and artificial intelligence techniques can also be incorporated to analyze classroom usage patterns and automatically optimize appliance control based on user behavior and environmental conditions. This would improve energy efficiency beyond fixed-threshold automation methods.

Additional environmental sensors such as humidity, air quality, and CO sensors can be integrated to create a healthier and more comfortable classroom atmosphere. The system can also be expanded to control more appliances such as projectors, air conditioners, and ventilation systems.

Future versions of the system may include cloud-based data storage and energy consumption analysis to help institutions monitor power usage and identify areas for improvement. Replacing mechanical relays

with solid-state relays can improve reliability and reduce maintenance requirements.

With these advancements, the system can evolve into a fully intelligent smart energy management platform capable of supporting modern sustainable buildings, educational institutions, offices, and other automated environments.

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APPENDIX

Arduino Code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// ———— PIN DEFINITIONS ————
#define PIR_PIN 14
#define LDR_PIN 34
#define LM35_PIN 32

#define RELAY_FAN 26
#define RELAY_LIGHT 25

#define FAN_SWITCH 33
#define LIGHT_SWITCH 27

// ———— OLED ————
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306
```

```
    display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// ---------- TIMING ----------
unsigned long lastUpdate = 0;
const int interval = 200;

// ---------- MOTION FILTER ----------
bool motionState = false;
unsigned long lastMotionTime = 0;
unsigned long lastHighTime = 0;

const unsigned long motionHoldTime = 5000;
const unsigned long lowConfirmTime = 500;

// ---------- MODES ----------
int fanMode = 0;    // 0=AUTO, 1=ON, 2=OFF
int lightMode = 0; // 0=AUTO, 1=ON, 2=OFF

bool lastFanSwitch = HIGH;
bool lastLightSwitch = HIGH;

// ---------- SETUP ----------
void setup() {

  Serial.begin(115200);
```

```
  pinMode(PIR_PIN, INPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LM35_PIN, INPUT);

  pinMode(RELAY_FAN, OUTPUT);
  pinMode(RELAY_LIGHT, OUTPUT);

  pinMode(FAN_SWITCH, INPUT_PULLUP);
  pinMode(LIGHT_SWITCH, INPUT_PULLUP);

  digitalWrite(RELAY_FAN, HIGH);
  digitalWrite(RELAY_LIGHT, HIGH);

  delay(3000);

  Wire.begin(21, 22);

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    while (true);
  }

  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);
}
```

```
// ---------- LOOP ----------
void loop() {

  if (millis() - lastUpdate >= interval) {
    lastUpdate = millis();

    // ---------- SWITCH HANDLING ----------
    bool fanSw = digitalRead(FAN_SWITCH);
    bool lightSw = digitalRead(LIGHT_SWITCH);

    if (lastFanSwitch == HIGH && fanSw == LOW) {
      fanMode++;
      if (fanMode > 2) fanMode = 0;
      delay(200);
    }

    if (lastLightSwitch == HIGH && lightSw == LOW) {
      lightMode++;
      if (lightMode > 2) lightMode = 0;
      delay(200);
    }

    lastFanSwitch = fanSw;
    lastLightSwitch = lightSw;

    int rawMotion = digitalRead(PIR_PIN);
```

```
// ———— MOTION FILTER ————
if (rawMotion == HIGH) {
  lastMotionTime = millis();
  lastHighTime = millis();
}

if (rawMotion == LOW) {
  if (millis() - lastHighTime > lowConfirmTime) {
    if (millis() - lastMotionTime > motionHoldTime) {
      motionState = false;
    }
  }
}

if (millis() - lastMotionTime <= motionHoldTime) {
  motionState = true;
}

int lightValue = analogRead(LDR_PIN);

int adcValue = analogRead(LM35_PIN);
float voltage = adcValue * (3.3 / 4095.0);
float temp = voltage * 100.0;

// ———— FAN CONTROL ————
```

```
if (fanMode == 1) {
  digitalWrite(RELAY_FAN, LOW); // ON
}
else if (fanMode == 2) {
  digitalWrite(RELAY_FAN, HIGH); // OFF
}
else {
  if (motionState && temp > 24) {
    digitalWrite(RELAY_FAN, LOW);
  } else {
    digitalWrite(RELAY_FAN, HIGH);
  }
}

// ---------- LIGHT CONTROL ----------
if (lightMode == 1) {
  digitalWrite(RELAY_LIGHT, LOW); // ON
}
else if (lightMode == 2) {
  digitalWrite(RELAY_LIGHT, HIGH); // OFF
}
else {
  if (motionState && lightValue >= 1600) {
    digitalWrite(RELAY_LIGHT, LOW);
  } else {
    digitalWrite(RELAY_LIGHT, HIGH);
```

```
    }
  }

  // ———————— OLED ————————
  display.clearDisplay();

  display.setCursor(0, 0);
  display.print("Temp: ");
  display.print(temp);
  display.println(" C");

  display.print("Motion: ");
  display.println(motionState ? "YES" : "NO");

  display.print("Light: ");
  display.println(lightValue);

  display.println("");

// FAN STATUS
if (fanMode == 1) display.println("FAN: MANUAL ON");
else if (fanMode == 2)
display.println("FAN: MANUAL OFF");
else display.println((motionState && temp > 24)
? "FAN: AUTO ON" : "FAN: AUTO OFF");
```

```
// LIGHT STATUS
if (lightMode == 1) display.println("LIGHT: MANUAL ON");
else if (lightMode == 2)
display.println("LIGHT: MANUAL OFF");
else display.println((motionState && lightValue >= 1600)
? "LIGHT: AUTO ON" : "LIGHT: AUTO OFF");

display.display();

// ---------- SERIAL ----------
Serial.print("FanMode: ");
Serial.print(fanMode);
Serial.print(" | LightMode: ");
Serial.print(lightMode);
Serial.print(" | Temp: ");
Serial.print(temp);
Serial.print(" | Motion: ");
Serial.println(motionState);
  }
}
endmodule
```